SDERI: Autonomous Navigation with Safe Subgoal Distributions and Exploration-Replanning Index in the Dynamic Environments

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***Abstract*— In crowded dynamic scenes, autonomous navigation must satisfy three key requirements: geometry safety, real-time performance, and interpretability. The mainstream pipelines typically consider only one or two requirements. Therefore, we propose SDERI, a hierarchical navigation framework that includes safe region subgoal distribution and exploration-replanning index (ERI). First, we bound the region of the subgoal distribution to a safety corridor that is consistent with global reachability. Then, an ERI is used to regulate the entropy of the subgoal distribution within the safe region and the replan timing of the current subgoal distribution. ERI is computed from minimal online cues, local occupancy ratio and normalized remaining distance. The physical simulation experiments demonstrate that our navigation method is effective in diverse dynamic scenarios. Meanwhile, it outperforms the baseline algorithms in terms of success rate and safety, especially in the case where the environment's crowd density increases. Moreover, the ablation experiments confirm the necessity of each component in this framework. A video of SDERI is hosted at https://sites.google.com/view/sderifornavigation.**

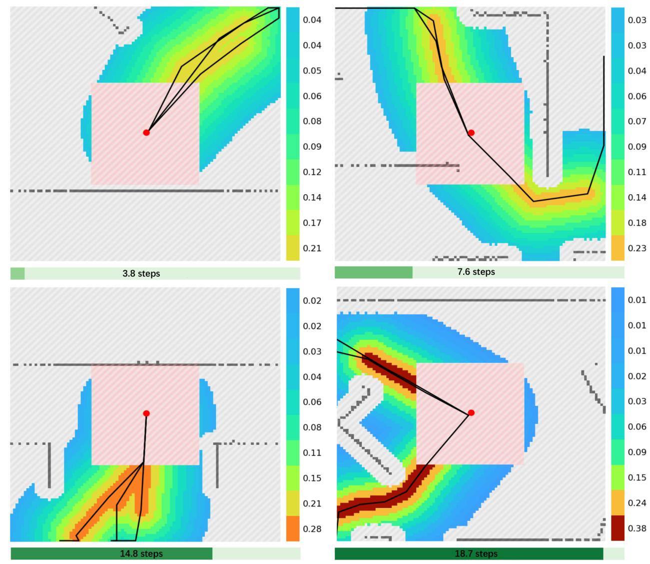

Fig. 1: Subgoal distributions and replanning times under different values of the exploration-replanning index (ERI), computed online from the local occupancy ratio and normalized remaining distance (Sec. IV). Panels are ordered left→right, top→bottom with ERI = $\{0.817, 0.603, 0.375, 0.129\}$. The background is a 100×100 local occupancy grid (dark gray area is obstacles); the black lines are the global paths generated by RRT that are reachability-equivalent; the red dot marks the robot, and the translucent red disc is the near-field exclusion zone that is removed from sampling. The colored heatmap shows the subgoal distribution restricted to the safe-region mask (with the probabilities of layers noted); the horizontal color band depicts the replanning time associated with the same ERI (darker = longer). As ERI increases, the distribution becomes more diffuse and the replanning time shortens; as ERI decreases, the distribution concentrates and the replanning time lengthens. This illustrates how a single ERI co-regulates dispersion and timing while the subgoal distribution stays within the safety-consistent corridor.

I. Introduction

Autonomous navigation is a crucial capability for robots in real-world settings. Highly dynamic scenes often face three concurrent requirements: (i) geometry safety with no collisions or boundary violations; (ii) real-time performance to unforeseen moving obstacles; and (iii) system interpretability. These requirements frequently conflict in practice: increasing responsiveness to dynamics can compromise trajectory stability and interpretability [1]. While overly conservative safety methods hinder the robot's ability to adapt to rapid environmental changes. To balance these demands, many pipelines favor hybrid navigation methods, which leverage the complementary strengths of different methods [2]. One kind of design is the hierarchical framework, where a global planner assigns subgoals or global paths, a local planner executes them, and replanning is triggered regularly or in response to specific events [3]–[11].

But there are problems with these pipelines that still need to be addressed. One is that within the above framework, there is no universal "best" time for replanning: overly frequent replanning induces oscillations, whereas infrequent replanning desynchronizes the agent from abrupt environmental changes [12]. Another problem is that common global planners often provide only a single intermediate waypoint (subgoal). When this waypoint falls into a temporarily blocked region, the local policy may be forced to wait or repeatedly replan, leading to lag and instability. To address this issue, some works construct a *subgoal distribution* for the local planner to sample from. For example, [13] uses a neural network to construct a subgoal distribution in real time but does not explicitly enforce safety constraints or provide clear interpretability; [14] uses a rule-based distribution with strong interpretability, but it does not explicitly account for dynamic obstacles, which can limit real-time performance in dynamic scenes.

Therefore, we introduce SDERI, a hierarchical navigation framework for dynamic environments, as shown in Fig. 2. SDERI combines two complementary ideas: (i) a safety-consistent corridor that restricts subgoals to reachable, low-risk free space near a global path, and (ii) an exploration-replanning index (ERI) that schedules exploration and replanning online within this feasible support.

To ensure safety and reachability before scheduling, we construct a corridor-shaped region around the global path

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using a distance field and exclude unsafe areas. This mask provides three benefits: (i) it separates "where sampling is feasible" from "how sampling is scheduled," (ii) it enforces that sampled subgoals are reachable and low-risk, reducing invalid or dangerous selections, and (iii) it yields an interpretable support whose geometry maps directly to the subgoal distribution shape, as shown in Fig. 1.

ERI is a lightweight difficulty proxy computed from two broadly available cues: a local occupancy ratio and a normalized remaining-distance measure. These cues are sufficient for ERI's role as a *scalar scheduler*: the local occupancy ratio captures how constrained the robot is now, while the remaining-distance captures how long the robot must sustain safe progress. When crowding increases or the remaining distance is large, ERI increases to broaden the subgoal distribution (higher entropy) and shorten the replanning time, encouraging faster resampling and more proactive exploration. When the scene stabilizes or the goal is near, ERI decreases to concentrate sampling and reduce unnecessary switches, improving trajectory smoothness. ERI is not intended to encode full spatial structure; feasibility is handled by the corridor mask, while ERI only modulates dispersion and commitment within feasible regions.

In summary, SDERI can balance the triple challenges in the following way: (i) safety: the safe region proveides a geometric feasibility constraint so sampled subgoals lie in low-risk free space; (iii) real-time performance: ERI controls the exploration degree and the replan timing, and it is changed in real time according to the surrounding context; (iii) Interpretability: how ERI regulates exploration and replanning and how the safe region is built are highly interpretable, making it stable and easy to adjust. Our contributions can be listed as follows:

- We propose SDERI, a hierarchical navigation framework that combines a safety-consistent subgoal support with ERI-driven scheduling to balance safety, responsiveness, and interpretability in dynamic environments.
- We introduce ERI, a lightweight and extensible scheduling index computed from minimal online cues, which co-regulates subgoal-distribution entropy and replanning time.
- We validate SDERI in multiple simulated dynamic scenes and on a real robot, comparing against different baselines in crowded, high-dynamic settings.

II. Related Work

A. Hierarchical Navigation and Subgoal Distributions

Hybrid global-local pipelines remain the popular choice for dynamic navigation [2], [15]–[17], with many works strengthening the global-local interface through explicit subgoals. For example, [4] connects Deep Reinforcement Learning based local avoidance to conventional global planners via waypoint generators; [18] generates adjacency-constrained subgoals that preserve connectivity; there are also hierarchical navigation from rough 2-D maps [19], [20] extracts multiple candidate points-of-interest and selecting an online waypoint, and human-aware subgoal generation to inject semantic cues [21]. More recently, several papers advocate subgoal distributions rather than single subgoals: learning attention-based subgoal distributions on real systems [13], successor representation-driven incremental subgoal discovery emphasizing multi-modality [14], jointly learning spatial subgoals and temporal choices [8], and formulating "where to go next" as a learnable recommendation policy [11]. These collectively suggest that a distributional supply adapts better to dynamic crowds than a single point; however, most do not explicitly confine the support to a geometry-consistent safe set, and thus struggle to optimize the triple challenge jointly.

B. When to Replan

In hybrid navigation, "when to replan" remains a less discussed but important issue: overly frequent replanning induces path oscillations, whereas infrequent replanning causes stalls or latency. Mainstream systems still rely on fixed periods or event thresholds [4], [6], [22]. Honda et al. provided the first systematic evaluation, showing that the optimal replanning timing is not universal across maps and global-local combinations. They cast the timing as a POMDP and trained a DRL replanning controller, thereby improving navigation in partially unknown environments [12]. Building on this, we replace the discrete on-off trigger with a single continuous scheduling index, improving interpretability and mitigating chattering.

C. Safety-Consistent Feasible Regions

There are two complementary threads that pursue safety under dynamics. A geometry-reachability-first line constructs spatio-temporal safety corridors and optimizes collision-free trajectories within them, e.g., RAST builds risk-aware corridors without strong assumptions on obstacle shapes and counts [23]. A probabilistic-safety line injects uncertainty into planning and control via conformal prediction (CP) and related filters, e.g., CP-based MPC with user-settable confidence levels [24], predictive safety filters layered over RL controllers to stay close to nominal behavior while avoiding high-uncertainty regions [25], and adaptive CP combined with constrained RL for social navigation [26]. Semantic cues can further specify class-dependent safety margins in crowded scenes [21]. Our work follows the geometry prior route by restricting the subgoal distribution to a safe region consistent with global reachability. Then, it uses ERI to co-regulate exploration entropy and commitment duration, aiming to improve both geometric safety and interpretable responsiveness simultaneously.

III. Navigation Framework

A. Problem Formulation

We consider a differential-drive robot with a limited sensor field of view that moves in a two-dimensional dynamic environment from an initial pose to a global goal within a specified time frame. The dynamic environment includes static obstacles and multiple independent dynamic entities. The task is naturally modeled as a partially observable

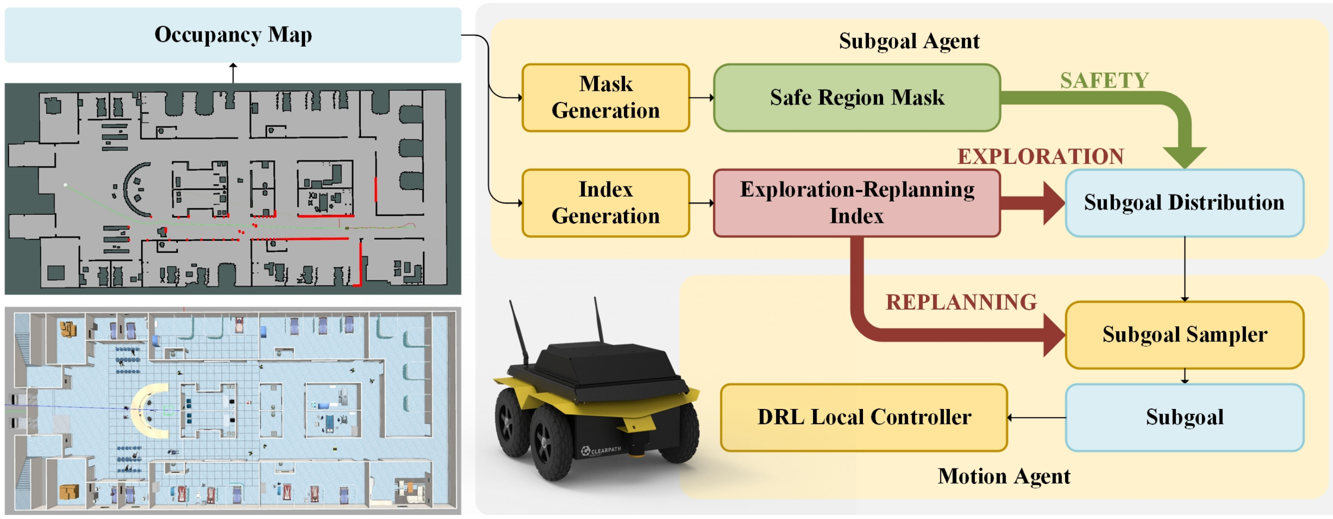


Fig. 2: Overview of SDERI. From the current occupancy grid, the Subgoal Agent builds a global-reachability-consistent safe-region mask and computes ERI. The mask bounds where subgoals may be sampled; ERI sets the distribution's entropy and a scene-adaptive replanning time. The Motion Agent samples a subgoal, tracks it with the DRL local controller, and triggers early replanning on subgoal-reached, stall, or timer-expiry events.

Markov decision process (POMDP). We separate decision making into two time scales: a macro-step index k with step size Δt (typically 0.5-1s), and a micro-step control loop running at 10-50 Hz. Each macro step spans a block of micro steps; at macro step k, an observation is made, that is, a local occupancy grid constructed online from a 2-D LiDAR.

B. Hierarchical Architecture

As illustrated in Fig.2, the system consists of two coupled layers: the Subgoal Agent and the Motion Agent. Given the current occupancy grid, the Subgoal Agent produces a safe-region mask $\mathcal{M}$ that is consistent with global reachability, and an ERI ξ that schedules the target entropy of the subgoal distribution and the replanning time. ERI is computed from minimal online cues and does not require explicit obstacle tracking, preserving portability and real-time performance.

Motion Agent consists of a subgoal sampler and a local end-to-end DRL controller. The sampler draws a subgoal from the distribution and computes a replanning time based on ERI; within this horizon, the controller tracks the subgoal and triggers resampling upon goal attainment, stalling, or expiration of the horizon.

IV. Mask and Index Generation

At macro step k, the Subgoal Agent produces two coupled artifacts: a safe-region mask $\mathcal{M}$ and an ERI ξ. The mask constrains subgoal samples to the inflated-free, reachability-consistent corridor in the current occupancy grid, while ERI schedules the dispersion of the subgoal distribution and the replanning time within this feasible support.

A. Safe Region Mask

The Subgoal Agent receives the current 2-D occupancy grid $\mathbf{O}_k \in \{0,1\}^{H\times W}$ at a macro time step k, and inflates obstacles by a radius $R = r_{\text{robot}} + s_{\text{margin}}$ to obtain a binary obstacle mask $\mathbf{B}_k$. Then, it computes M global reference paths $\{\Gamma_m\}_{m=1}^{M}$ using RRT [27]. The paths are intentionally diversified yet reachability-equivalent, to widen the safe support for subgoal sampling. Around the path set, the agent forms an Euclidean distance field $D(x)$:

$$D(x) = \min_{m} \min_{p\in\Gamma_m} \|x-p\|_2, \tag{1}$$

where x denotes a candidate cell in the environment, and $p \in \Gamma_m$ is a waypoint on that path. Thus, $D(x)$ measures the shortest Euclidean distance from x to any point along any reference path. Based on it, we define a safety corridor $\Omega_\rho = \{x \mid D(x) \le \rho\}$ so that it is consistent with global reachability, where ρ is the maximum safety distance. To limit computation, we discretize Ω_ρ into N thin layers of equal width $\Delta\rho = \rho/N$. The layer j collects all cells with

$$(j-1)\Delta\rho < D(x) \le j\Delta\rho. \tag{2}$$

Cells inside the same layer are treated as equivalent for later distribution allocation (Sec. V-A). Finally, a small disc around the robot is excluded from sampling (visualized as the red semi-transparent region in Fig. 1), preventing sampling the subgoals too close to the robot itself. The near-field exclusion is implemented on the occupancy grid and aligned with the inflated robot footprint; due to discretization it appears square in the visualization. In conclusion, the safe-region mask $\mathcal{M}$ is the union of layer cells that are inside Ω_ρ, outside $\mathbf{B}_k$ after inflation, and outside the near-field exclusion zone.

Safety scope and limitations. The safe-region mask $\mathcal{M}$ provides a support-level safety prior: it constrains subgoal samples to the inflated-free, reachability-consistent corridor defined on the current occupancy grid. However, this does not constitute a formal guarantee of collision-free execution. Safety during subgoal tracking still depends on the local controller. Accordingly, to prove the mask still improves overall navigation safety, we report safety using SR/AC in Sec. VI.

B. Exploration-Replanning Index

The ERI $\xi \in [0, 1]$ is a lightweight scheduling index for dynamic navigation. It is not a full scene descriptor; instead, it summarizes (i) instantaneous local interaction pressure and (ii) remaining task horizon to adapt exploration and commitment online. We increase it when local crowding rises or when the goal lies farther away (more dispersion, shorter commitment). We decrease it when the scene stabilizes or the goal nears (tighter distribution, longer commitment). Here's how ERI is calculated.

After building the safe region mask, the Subgoal Agent computes two features at the macro step k: the crowding score ρ_k and the normalized remaining distance d_k.

Let $O_k(c) \in \{0, 1\}$ indicate whether cell c in the local occupancy grid is occupied. Then, define a circular window centered at the current pose $\mathbf{x}_k$ with radius R_p,

$$S_k = \{\, c \in \Omega_k \mid \|\mathbf{p}(c) - \mathbf{x}_k\| \le R_p \,\}, \tag{3}$$

where $\mathbf{p}(c)$ is the geometric center of cell c. And the crowding score is the occupied-cell ratio in the window

$$\rho_k = \frac{1}{|S_k|} \sum_{c \in S_k} O_k(c). \tag{4}$$

Let L_{tot} denote the arc length of the shortest path in global paths $\Gamma = \{\Gamma_m\}_{m=1}^{M}$. Let s_k be the arc-length parameter of the orthogonal projection of $\mathbf{x}_k$ onto that shortest path. The remaining path length is

$$L_{\text{rem}}(k) = L_{\text{tot}} - s_k \ge 0 \tag{5}$$

We normalize it to $[0, 1]$, with a small $\varepsilon > 0$ to avoid division by zero

$$d_k = \min\Big(1,\ \frac{L_{\text{rem}}(k)}{L_{\text{tot}} + \varepsilon}\Big) \in [0, 1]. \tag{6}$$

We blend the two features with a weight $\alpha \in [0, 1]$ and bound the result within user-set limits $\xi_{\min} \le \xi_{\max}$ as follows:

$$\begin{aligned} \tilde{\xi}_k &= \alpha\, \rho_k + (1 - \alpha)\, d_k \\ \xi_k &= \xi_{\min} + (\xi_{\max} - \xi_{\min})\, \tilde{\xi}_k. \end{aligned} \tag{7}$$

In our experiments, the ERI parameters, α, R_p, $\xi_{\min}$, $\xi_{\max}$ are fixed across scenes to avoid per-scenario hand tuning. Thus, higher crowding ($\rho_k \uparrow$) or a farther goal ($d_k \uparrow$) increases ξ_k linearly, raising distribution entropy and shortening replanning time; spacious scenes or near-goal conditions lower ξ_k, shrinking the distribution and extending replanning time. From ξ_k the Subgoal Agent provides: (a) a target entropy level to regulate the subgoal distribution within $\mathcal{M}$ (Sec. V-A), and (b) a replanning time for the Motion Agent (Sec. V-B), which will be mapped from ξ_k.

Why these two features? ERI only needs to support scheduling decisions, which are how exploratory to be and how often to replan, not to represent full geometry or predict obstacle motion. For scheduling, two signals are sufficient: (i) *local interaction pressure* ρ_k, indicating how constrained the robot is by nearby occupied space, including both static obstacles and dynamic agents reflected in the occupancy grid; and (ii) *remaining horizon* d_k, indicating how long the robot must sustain safe progress before termination. Spatial structure is handled separately by the safety-consistent corridor mask. ERI only modulates dispersion and commitment within this feasible set. We use a bounded monotone blend to keep the scheduler stable and interpretable; other monotone maps or additional normalized terms (e.g., velocity/flow estimates) can be incorporated when reliable tracking is available, but are not required for SDERI to function.

V. Subgoal Distribution and Subgoal Sampler

A. Subgoal Distribution

Given the safe-region mask $\mathcal{M}$, we partition it into N thin layers (near-to-far from the global-path set as in Sec. IV-A). Cells in the same layer are treated as equivalent for allocation; we then sample uniformly within each layer to approximate a continuous corridor distribution with very low overhead. Let $\mathcal{B}_j \subset \mathcal{M}$ be the layer $j \in \{1, \ldots, N\}$ and $n_j = |\mathcal{B}_j|$ its cell count. We use the ERI ξ_k to control dispersion:

$$T(\xi_k) = T_{\min} + (T_{\max} - T_{\min})\, \xi_k,\ 0 < T_{\min} < T_{\max} \tag{8}$$

Larger ξ_k gives larger T, flattening layer preferences (more exploratory), while smaller ξ_k concentrates probability on inner layers (more conservative). We choose a bounded, monotone map from ξ_k to T so that increasing scene difficulty cannot reduce exploration, and so exploration remains within calibrated limits $[T_{\min}, T_{\max}]$ for stable behavior. The linear form is used for simplicity and interpretability; any bounded monotone mapping would be compatible with SDERI. In our evaluation, $T_{\min}$ and $T_{\max}$ are fixed across all scenes.

We score layers with a distance-preference term and a capacity adjustment for unequal layer sizes:

$$\ell_j = -\frac{j}{T(\xi_k)} \ +\ \beta \log\big(n_j + \varepsilon_{\text{layer}}\big),\ \beta \in [0, 1], \tag{9}$$

$$q_j = \frac{\exp(\ell_j)}{\sum_{i=1}^{N} \exp(\ell_i)}. \tag{10}$$

The first term encodes that farther layers are less preferred as T decreases. The second term compensates for unequal numbers of cells across layers. After uniform sampling within the layer, each cell's probability becomes insensitive to n_j. Thus forms the subgoal distribution

$$p(c) = \frac{q_j}{n_j}, \quad c \in \mathcal{B}_j,\ \sum_{c \in \mathcal{M}} p(c) = 1. \tag{11}$$

Equations (8)-(10) define a Gibbs distribution over layer index j. As T increases, the weights q_j become flatter; as T decreases, mass concentrates on small j. The Shannon entropy of q therefore increases monotonically with T. This procedure maintains support strictly within the safety region

Fig. 3: Benchmark scenes in Arena-3D (left→right, top→bottom): Scene 1, map5 gazebo (sparse, structured layout); Scene 2, factory gazebo (large industrial floor with bulky static occluders); Scene 3, aws house gazebo (multi-room indoor environment with narrow passages); Scene 4, small warehouse gazebo (compact, crowded warehouse, the most challenging one). The four scenes form an increasing gradient of geometric and dynamic difficulty used throughout the evaluation.

generated in Sec. IV-A, and adapts dispersion via ξ on a single scale.

B. Subgoal Sampler

Given the subgoal distribution $p(\cdot)$ (Sec. V-A) and the ERI ξ_k (Sec. IV-B), the sampler chooses a concrete subgoal. It sets a replanning time that determines the latest moment the system will query the Subgoal Agent again.

Let δ be the control period of the micro-loop (e.g., $20\,\mathrm{Hz} \Rightarrow \delta = 0.05\,\mathrm{s}$). We select a replanning time,

$$\begin{aligned} t_k^{\text{replan}} &= t_{\min} + (t_{\max} - t_{\min})\, s(1 - \xi_k), \\ s(z) &= z^{\gamma}, \quad \gamma \geq 1, \end{aligned} \tag{12}$$

where $0 < t_{\min} \leq t_{\max}$. We map ERI to replanning time using a bounded, monotone decreasing function of ξ_k to prevent pathological behaviors: excessively frequent replanning in hard scenes and overly long commitments in dynamic interactions. Specifically, larger ξ_k yields shorter replanning time, while smaller ξ_k yields longer replanning time. The exponent $\gamma \geq 1$ controls sensitivity: it keeps the schedule smooth and biases toward longer commitments when the scene is stable. The bounds $t_{\min}$ and $t_{\max}$ are determined by control rate and compute budget constraints, and are fixed across scenes to avoid per-scenario hand tuning.

$$\begin{aligned} H_k &= \operatorname{clip}\left(\left\lfloor \frac{t_k^{\text{replan}}}{\delta} \right\rfloor,\, H_{\min},\, H_{\max}\right), \\ H_{\min} &= \lceil t_{\min}/\delta \rceil\,,\ H_{\max} = \lfloor t_{\max}/\delta \rfloor\,. \end{aligned} \tag{13}$$

Starting from the current robot pose $\mathbf{x}_t$ and subgoal $\mathbf{g}_k$, the local controller tracks $\mathbf{g}_k$ for at most H_k control steps. The system replans immediately when the subgoal is reached ($\|\mathbf{x}_t - \mathbf{g}_k\| \leq \varepsilon_{\text{goal}}$) or the H_k steps elapse (i.e., the replanning time is reached). Upon any trigger, control returns to the Subgoal Agent to rebuild $\mathcal{M}$, recompute ξ, update $p(\cdot)$, and draw a new subgoal.

VI. Experiments

A. Experimental Setup

1) Simulation Platform and Robot Model: All simulated experiments are conducted on an x86_64 Ubuntu workstation equipped with an Intel® Core™ i5-10400F CPU @ 2.90 GHz (12 cores), an NVIDIA GeForce RTX 3080 GPU, and 64 GB RAM. We use Gazebo as the simulator. Environment maps are represented as 2-D occupancy grids with a resolution of 0.05 m/cell. The test robot is Clearpath Jackal with a differential drive. For collision checking, the footprint is approximated by a rectangle of 0.42 m × 0.33 m. Perception uses a 2-D LiDAR with a 360° field of view and a 5 m maximum range. The control stack runs at two time scales: a low-level control loop at 20 Hz, and a macro-decision cycle of 1 s, which is the update rate for ERI.

2) Baselines: To evaluate SDERI, we compare it against four navigation baselines: (i) DWA [17]; (ii) MPC [15], [28]; (iii) MACA, a classic end-to-end deep reinforcement learning navigator [29]; (iv) GDAE, a hierarchical navigator with a high-level DQN selects intermediate subgoals from a discrete direction-distance set [20]. All methods share the

TABLE I: RESULTS ACROSS METHODS AND SCENES

Scene	Method	SR ↑	AT ↓	AC ↓
Scene 1	DWA	0.950	52.93	0.656
	MPC	0.941	63.56	0.960
	MACA	0.970	31.37	0.254
	GDAE	0.950	43.50	0.420
	SDERI	**0.990**	**20.83**	**0.166**
Scene 2	DWA	0.960	**45.42**	0.549
	MPC	0.818	92.88	1.212
	MACA	0.980	48.64	**0.107**
	GDAE	**1.000**	64.41	0.290
	SDERI	0.990	48.38	0.137
Scene 3	DWA	0.934	13.70	0.881
	MPC	**1.000**	40.48	0.114
	MACA	**1.000**	**10.24**	0.099
	GDAE	**1.000**	10.71	**0.060**
	SDERI	**1.000**	11.49	0.085
Scene 4	DWA	0.941	26.89	0.686
	MPC	0.735	440.10	1.696
	MACA	0.990	21.23	0.127
	GDAE	0.990	17.36	0.280
	SDERI	**1.000**	**15.24**	**0.098**
Avg.	DWA	0.946	34.73	0.693
	MPC	0.873	159.26	0.995
	MACA	0.985	27.87	0.147
	GDAE	0.985	34.00	0.263
	SDERI	**0.995**	**23.99**	**0.121**

TABLE II: THE RESULT OF THE ABLATION STUDY

Exp.	Com.	Saf.	SR ↑	AT ↓	AC ↓
✗	✗	✗	0.98	23.02	0.32
✗	✓	✓	**1**	14.76	0.16
✓	✗	✓	**1**	**14.57**	0.11
✓	✓	✗	0.99	20.26	0.26
✓	✓	✓	**1**	15.27	**0.10**

same settings. We used MACA as the local controller of GDAE and SDERI framework in the experiments.

3) Scenes and Tasks: We evaluate in four simulated scenes in Arena-rosnav-3D [30], [31] to span a rising gradient of spatial and dynamic complexity, as in Fig. 3.

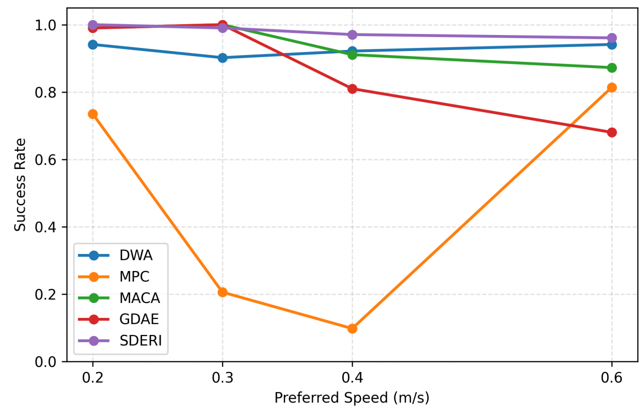
- Scene 1 is map5 gazebo (sparse baseline; 10 pedestrians). Open, simple, structured layout with few occluders. Pedestrian interactions are light.
- Scene 2 is factory gazebo (structured industrial; 10 pedestrians). Large factory floor with large, static obstacles forming aisles and cavities.
- Scene 3 is aws house gazebo (indoor housing; 5 pedestrians). Multi-room indoor environment with narrow passages and many small static obstacles. Frequent doorways and turns create tight clearances and visibility changes.
- Scene 4 is small warehouse gazebo (compact and crowded; 10 pedestrians). Space-limited warehouse with dense shelving and sustained human traffic. The most challenging case, combining tight geometry, occlusions, and crowding.

Each episode is a point-goal navigation problem. We ran 100 episodes for each baseline and each scene. Pedestrians are generated by the Arena-rosnav-3D crowd modules for each scene; their trajectories vary across episodes.

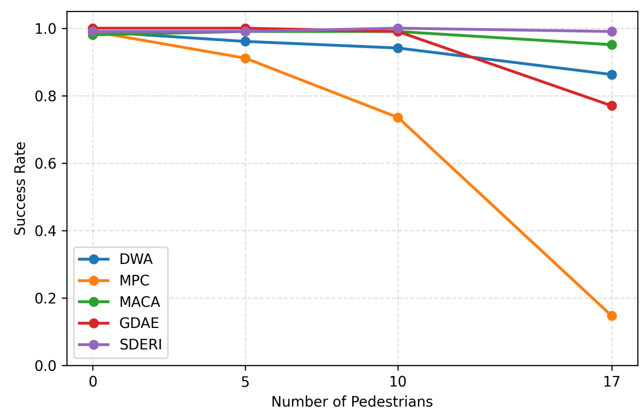
4) Evaluation Metrics: We report the following quantitative metrics:

TABLE III: Feature ablation on Scene 4.

ERI features	SR↑	AT (s)↓	AC↓
base (τ, ρ)	**1.00**	**15.24**	**0.10**
base + ρ_{front}	0.99	21.29	0.31
base + b_{corr}	1.00	22.59	0.34
base + ρ_{front} + b_{corr}	0.83	73.47	1.13



(a) Success rate vs. pedestrian speed.



(b) Success rate vs. pedestrian count.

Fig. 4: Robustness under varying crowd dynamics: (a) success rate versus pedestrian speed; (b) success rate versus pedestrian count.

- Success Rate (SR, %) — Fraction of episodes in which the robot reaches the goal safely and with terminal pose error below a preset threshold. As the default setting in Arena, we count an episode as safe if transient overlaps occur at most three times during the run.
- Average Time (AT, s) — Mean time from start to goal over successful episode.
- Average Collisions (AC) — Mean number of collisions per episode. Unlike SR, we use a strict definition of collisions on AC. A collision is recorded whenever the (inflated) obstacle region overlaps the (inflated) robot footprint.

B. Results and Discussions

1) Results in Different Scenes: Table I reports performance across four scenes. SDERI achieves the best overall average. Compared with GDAE, SDERI reduces AT by

TABLE IV: PER-EPISODE REAL-WORLD RESULTS OVER 10 RUNS (7 PEDESTRIANS, $5\,\mathrm{m} \times 5\,\mathrm{m}$).

Method	Metric	1	2	3	4	5	6	7	8	9	10	Avg.
TEB	Time (s)	48	40	64	34	48	50	47	47	50	51	47.9
	Stuck	1	0	0	0	1	0	1	1	0	0	0.4
	Jerk	2	2	3	2	2	3	2	2	3	4	2.5
SDERI	Time (s)	26	33	29	34	28	35	26	35	24	37	**30.7**
	Stuck	0	0	0	0	0	0	0	0	0	0	**0.0**
	Jerk	1	0	0	0	0	1	1	0	0	0	**0.3**

Fig. 5: The real-world experiment with Autolabor Pro1 robot.

29.4% and AC by 54.0%, while improving SR from 0.985 to 0.995.

Scene-wise, SDERI remains consistently strong across the difficulty gradient. In Scene 1, SDERI leads on all metrics. In Scene 2, SDERI maintains near-identical time to MACA with high SR and substantially lower AC than other baselines. In Scene 3, all learning methods reach SR=1.000; SDERI stays competitive. In the most challenging Scene 4, SDERI dominates, outperforming all other baselines.

Overall, GDAE improves robustnes in several cases, but tends to trade efficiency and/or safety in dense clutter and crowds. SDERI closes this gap by enforcing reachability-consistent subgoal support and adapting exploration and replanning via ERI, which is most beneficial in the hardest scenes.

2) Results under Varying Crowd Dynamics: Fig. 4a–4b evaluate Scene 4 under increasing pedestrian speed and density. SDERI shows the least degradation as dynamics intensify: success remains near-saturated across the sweeps, while baseline methods exhibit clearer sensitivity to faster and denser crowds. Robustness curves show SDERI's advantage growing with difficulty.

3) Ablation Study: We decompose the framework into three switchable components and test them on the same simulation task in Scene 4 (Table II):

- Exp.: ERI regulates the entropy of the subgoal distribution (✓) vs. fixed entropy (✗).
- Com.: ERI regulates the replan time (✓) vs. a fixed replan time (✗).
- Saf.: subgoals are constrained to a safe region (✓) vs. unconstrained support (✗).

The full model achieves the best safety with near-optimal efficiency. Removing the safe-region constraint (✓/✓/✗) substantially increases collisions and slows navigation, indicating that reachability-consistent support is critical. With Saf. enabled, entropy control alone (✓/✗/✓) yields the fastest AT and low AC, while replanning-only (✗/✓/✓) is slower and less safe. Joint control (✓/✓/✓) achieves the lowest AC, showing that combining dispersion shaping with adaptive replanning best stabilizes behavior in dense, high-dynamic crowds.

4) Ablation Study on ERI Features: We ablate ERI's feature space (τ, ρ) with two additional handcrafted cues: a goal-directed front-sector crowding ratio ρ_{front} and a corridor occupancy ratio b_{corr}. On Scene 4, as shown in Table III, the original feature set achieves the best overall performance. Adding either ρ_{front} or b_{corr} increases traversal time and collision rate significantly, and combining both degrades performance substantially, with long detours and unstable motion. Adding ρ_{front} and b_{corr} likely overestimates difficulty, making ERI too aggressive and causing long detours, more collisions, and more failures. These results suggest that (τ, ρ) already provides a sufficient difficulty proxy for scheduling, while additional handcrafted terms can introduce redundancy and variance that harms subgoal selection.

C. Real-World Validation

As shown in Fig 5, we conducted real-world experiments to validate the feasibility of SDERI under multi-pedestrian interactions. The robot, an Autolabor Pro1 differential-drive mobile base equipped with Mid360 lidar, navigated in a $5\,\mathrm{m} \times 5\,\mathrm{m}$ indoor space with 7 pedestrians moving at random speeds. To validate its feasibility, we compared SDERI against a high-frequency TEB planner. Each method was evaluated over 10 end-to-end episodes with identical start/goal configurations. All runs were successful and no collisions were observed for either method. We logged (i) traversal time, (ii) the number of stuck events, and (iii) the number of jerk events. As summarized in Table IV, SDERI reduced the average traversal time, achieved zero stuck events, and substantially reduced jerk events. Representative real-world runs are included in the supplementary video.

VII. CONCLUSION

We proposed SDERI, a hierarchical navigation framework that couples a reachability-consistent safe-region subgoal support with an exploration-replanning index (ERI) that co-regulates subgoal-distribution entropy and replanning timing from minimal online cues. Across Arena-rosnav-3D scenes, SDERI achieves the best average performance, and ablations confirm the complementary roles of its components; we also demonstrate feasibility on an Autolabor Pro1 platform. However, the limitations remain: the safe-region mask is a support-level, grid-based feasibility prior and does not guarantee collision-free execution, which still depends on the downstream controller and perception under dynamic interactions. Future work includes learning scene-aware yet interpretable ERI mappings, and adding safety filters or verification layers to strengthen execution-time guarantees.

REFERENCES

[1] C. Mavrogiannis, F. Baldini, A. Wang, D. Zhao, P. Trautman, A. Steinfeld, and J. Oh, "Core challenges of social robot navigation: A survey," 2021. [Online]. Available: https://arxiv.org/abs/2103.05668

[2] M. Shanmugaraja, M. Thangamuthu, and S. Ganesan, "Hybrid path planning algorithm for autonomous mobile robots: A comprehensive review," *Journal of Sensor and Actuator Networks*, vol. 14, no. 5, p. 87, 2025.

[3] W. Zhu and M. Hayashibe, "A hierarchical deep reinforcement learning framework with high efficiency and generalization for fast and safe navigation," *IEEE Transactions on Industrial Electronics*, vol. 70, no. 5, pp. 4962–4971, 2022.

[4] L. Kästner, X. Zhao, T. Buiyan, J. Li, Z. Shen, J. Lambrecht, and C. Marx, "Connecting deep-reinforcement-learning-based obstacle avoidance with conventional global planners using waypoint generators," in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2021, pp. 1213–1220.

[5] D. G. Eladio, M. Luis, and N. Angela, "Dynamic sub-goal generation with uncertainty analysis for mobile robots navigation," in *Submit to the 6th World Multi-conference on SYSTEMICS, CYBERNETICS AND INFORMATICS SCI*, 2002, pp. 14–18.

[6] K. Ota, Y. Sasaki, D. K. Jha, Y. Yoshiyasu, and A. Kanezaki, "Efficient exploration in constrained environments with goal-oriented reference path," in *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2020, pp. 6061–6068.

[7] G. J. Stein, C. Bradley, and N. Roy, "Learning over subgoals for efficient navigation of structured, unknown environments," in *Conference on Robot Learning*. PMLR, 2018, pp. 213–222.

[8] X. Huang, G. Sóti, C. Ledermann, B. Hein, and T. Kröger, "Planning with learned subgoals selected by temporal information," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 9306–9312.

[9] M. Bortkiewicz, J. Łyskawa, P. Wawrzyński, M. Ostaszewski, A. Grudkowski, B. Sobieski, and T. Trzciński, "Subgoal Reachability in Goal Conditioned Hierarchical Reinforcement Learning," in *16th International Conference on Agents and Artificial Intelligence*. SCITEPRESS, 2024.

[10] S. Pateria, B. Subagdja, A.-H. Tan, and C. Quek, "Value-Based Subgoal Discovery and Path Planning for Reaching Long-Horizon Goals," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 8, pp. 10 288–10 300, Aug. 2024.

[11] B. Brito, M. Everett, J. P. How, and J. Alonso-Mora, "Where to go next: Learning a subgoal recommendation policy for navigation in dynamic environments," *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 4616–4623, 2021.

[12] K. Honda, R. Yonetani, M. Nishimura, and T. Kozuno, "When to replan? an adaptive replanning strategy for autonomous navigation using deep reinforcement learning," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 6650–6656.

[13] J. De Heuvel, W. Shi, X. Zeng, and M. Bennewitz, "Subgoal-driven navigation in dynamic environments using attention-based deep reinforcement learning," in *2023 21st International Conference on Advanced Robotics (ICAR)*. IEEE, 2023, pp. 79–85.

[14] R. K. Dubey, S. S. Sohn, J. Abualdenien, T. Thrash, C. Hoelscher, A. Borrmann, and M. Kapadia, "SNAP:successor entropy based incremental subgoal discovery for adaptive navigation," in *Motion, Interaction and Games*. Virtual Event Switzerland: ACM, Nov. 2021, pp. 1–11.

[15] C. Roesmann, W. Feiten, T. Woesch, F. Hoffmann, and T. Bertram, "Trajectory modification considering dynamic constraints of autonomous robots," in *ROBOTIK 2012; 7th German Conference on Robotics*, May 2012, pp. 1–6.

[16] A. Faust, K. Oslund, O. Ramirez, A. Francis, L. Tapia, M. Fiser, and J. Davidson, "Prm-rl: Long-range robotic navigation tasks by combining reinforcement learning and sampling-based planning," in *2018 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2018, pp. 5113–5120.

[17] D. Fox, W. Burgard, and S. Thrun, "The dynamic window approach to collision avoidance," *IEEE Robotics and Automation Magazine*, vol. 4, no. 1, pp. 23–33, 2002.

[18] T. Zhang, S. Guo, T. Tan, X. Hu, and F. Chen, "Generating adjacency-constrained subgoals in hierarchical reinforcement learning," *Advances in Neural Information Processing Systems*, vol. 33, pp. 21 579–21 590, 2020.

[19] C. Xu, C. Amato, and L. L. S. Wong, "Hierarchical robot navigation in novel environments using rough 2-D maps," Jun. 2021.

[20] R. Cimurs, I. H. Suh, and J. H. Lee, "Goal-driven autonomous exploration through deep reinforcement learning," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 730–737, 2022.

[21] L. Kästner, J. Lil, Z. Shen, and J. Lambrecht, "Enhancing navigational safety in crowded environments using semantic-deep-reinforcement-learning-based navigation," in *2022 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*. IEEE, 2022, pp. 87–93.

[22] O. Brock and O. Khatib, "High-speed navigation using the global dynamic window approach," in *Proceedings 1999 Ieee International Conference on Robotics and Automation (Cat. No. 99CH36288C)*, vol. 1. IEEE, 1999, pp. 341–346.

[23] G. Chen, S. Wu, M. Shi, W. Dong, H. Zhu, and J. Alonso-Mora, "Rast: Risk-aware spatio-temporal safety corridors for mav navigation in dynamic uncertain environments," *IEEE Robotics and Automation Letters*, vol. 8, no. 2, pp. 808–815, 2022.

[24] L. Lindemann, M. Cleaveland, G. Shim, and G. J. Pappas, "Safe planning in dynamic environments using conformal prediction," *IEEE Robotics and Automation Letters*, vol. 8, no. 8, pp. 5116–5123, 2023.

[25] K. J. Strawn, N. Ayanian, and L. Lindemann, "Conformal predictive safety filter for rl controllers in dynamic environments," *IEEE Robotics and Automation Letters*, 2023.

[26] J. Yao, X. Zhang, Y. Xia, Z. Wang, A. K. Roy-Chowdhury, and J. Li, "SoNIC: Safe social navigation with adaptive conformal inference and constrained reinforcement learning," Feb. 2025.

[27] J. Wang and E. Zheng, "Path planning of a mobile robot based on the improved rapidly exploring random trees star algorithm," *Electronics*, vol. 13, no. 12, p. 2340, Jan. 2024.

[28] C. Rösmann, A. Makarow, and T. Bertram, "Online motion planning based on nonlinear model predictive control with non-euclidean rotation groups," in *2021 European Control Conference (ECC)*, Jun. 2021, pp. 1583–1590.

[29] P. Long, T. Fan, X. Liao, W. Liu, H. Zhang, and J. Pan, "Towards optimally decentralized multi-robot collision avoidance via deep reinforcement learning," in *2018 IEEE International Conference on Robotics and Automation (ICRA)*, May 2018, pp. 6252–6259.

[30] L. Kästner, T. Bhuiyan, T. A. Le, E. Treis, J. Cox, B. Meinardus, J. Kmiecik, R. Carstens, D. Pichel, B. Fatloun, N. Khorsandi, and J. Lambrecht, "Arena-bench: A benchmarking suite for obstacle avoidance approaches in highly dynamic environments," 2022. [Online]. Available: https://arxiv.org/abs/2206.05728

[31] L. Kästner, T. Buiyan, L. Jiao, T. A. Le, X. Zhao, Z. Shen, and J. Lambrecht, "Arena-rosnav: Towards deployment of deep-reinforcement-learning-based obstacle avoidance into conventional autonomous navigation systems," in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, pp. 6456–6463.